

Paper title: Agent-Based Interactions and Economic Encounters in an Intelligent InterCloud

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Coalition formation and payoff division method:

The paper focuses on creating mathematical foundations for InterCloud economics and novel interaction protocols and negotiation strategies for agent-based InterCloud coalition formation and payoff division. The InterCloud is a “cloud of clouds” where clouds (agents) cooperate to share resources with each other so they can best satisfy consumer requests. Therefore, the sub-component interactions of the InterCloud system involve both cloud-to-cloud (provider-provider) interactions and cloud-stakeholder (provider-consumer) interactions where autonomous agents represent both individual clouds and stakeholders, the InterCloud being modeled as a multi-agent system.

Both a four-step coalition formation protocol between agents in this system and subgame perfect equilibrium strategies that result in fair payoff division (equal to each agent’s Shapley value) are introduced. Since the paper introduces a complete economic model, a section on a proposed adaptive concession rate-minimally sufficient concession (ACR-MS) negotiation strategy for InterCloud is also included but I will focus on summarizing coalition and payoff division methods for the sake of this assignment.

In the economic model of the InterCloud, coalition formation may occur between clouds when a “home” cloud need help resource capacity from some “foreign” clouds, in order to meet service requests from a consumer. In this case, the home cloud will need to ‘purchase’ resource capacity from other clouds via a service-level agreement (SLA) including the price and time slot for resource usage. Home cloud agents (hCAs) then negotiate with foreign cloud agents (fCAs) to form InterCloud coalitions. The 4-step formation protocol follows:

The first step is a form of “announcement” where every cloud that has any resource capacity available but not sufficient alone to provide a service broadcasts a “resource available” message to all other cloud agents in the InterCloud, specifying capacity amount and availability time duration.

Then the second step is a “preliminary contract” where each hCA that needs more resource capacity will select a subset of fCAs that have the appropriate resource type and amount of resource capacity. A sealed bid profile is then made by the hCA to the members of the chosen fCA subset to bid for the *right to acquire* every member’s resource capacity. It is proven that the best bid that can be made by a hCA_i to each fCA in its desired group is the Shapley value that fCA would have gotten in coalitions without the hCA subtracted from the Shapley value it would have gotten in coalition including the hCA_i . Each fCA can receive bids from more than one hCA, and the highest bidder hCA wins the right to acquire the resource capacity of some fCA. Only if a hCA wins right to acquire resource capacity of every fCA in its requested subset does it proceed to step three, otherwise it drops out with a payoff of 0.

This step is performed to avoid a situation where a coalition more profitable to all members is being formed because a part of the required resources are held by another proposer who wants to form another coalition.

Step three is a “contract offer” in which the highest bidder hCA makes a proposal profile for the members of its desired subset of fCAs. The stated best proposal strategy is for the hCA (surviving from step two) to offer each fCA in its desired subset the Shapley value it would have gotten in games not involving the hCA, as that is the minimum the fCA will accept.

Then the fourth step is an “acceptance/rejection” process where every fCA accepts and rejects payments proposed to it. The proven best strategy is for every fCA in a proposed coalition to accept a proposal that is equal or greater than the Shapley value it would have gotten in games (of the current proposed coalition subset) not involving the bidding hCA and to reject all proposals lower than it.

It is proven in the paper that in a zero-monotonic InterCloud coalition game, the subgame perfect equilibrium outcome of the 4-stage protocol coincides with the payoff vector prescribed by the Shapley value.

Issues raised on payoff division:

The paper provides a framework for the coalition formation and payoff division of an agent-based InterCloud showing optimality, stability, and fairness rigorously through a series of mathematical proofs. However, because the optimality of this method is only shown from mathematical requirements, performance issues or implementation difficulties are unknown, as no experiment has been done on an actual version of this coalition method. Additionally, the coalition framework relies on a rather specific and unique characterization of InterCloud interaction, with respect to “home” and “foreign” agents. Possibly due to the novelty of this work, this has not been mentioned in previous works or discussion about InterCloud. Since the method has not been tested in reality, it is hard to know for sure if this type of characterization of motivation for coalition formation is too specific for InterCloud interaction, as other types of cloud-to-cloud coalition interactions can be possible but are not explored by this mathematical model.

There are small handwavy parts to the method described, such as in stage 1 of the coalition formation protocol where hCAs must select a subset of fCAs with “appropriate” resource types and amounts of capacity, but these qualities are not defined or described at all, even though their ease of characterization and identification can heavily influence any actual implementation of this protocol.

Some parts of the coalition protocol seem very inflexible in terms of catering to conditions in reality, for example a hCA has to drop out of the game if it cannot essentially be the highest bidder for every single fCA requested for the resource capacity boost. If we want the success rate for something like this in reality to be higher, where there can be some slack between “ideal” and “acceptable” resource allocation amount it seems like some flexibility should be introduced into the algorithm so that a hCA can still get sufficient amount of needed resources such as through reaching some minimal ratio of successful bids for its requested fCAs.

Shapley value is also used in every payoff division strategy for the protocol and it takes factorial time to compute, which is not considered fast by any means.